Lecture Scribe: Lecture 14 – Tutorial-2 Solutions

Course: CSE 400: Fundamentals of Probability in Computing

February 20, 2026

Topic: Comprehensive Solutions to Probability Problem Set 2

Q1. Geometric Probability: Ratio of Segments

Problem: A point is chosen at random on a line segment of length L. Find the probability that the ratio of the shorter to the longer segment is less than 1/4.

Reasoning & Logic

- **Uniform Selection:** "At random" implies the point x is chosen from a uniform distribution over $(0, L)$.
- **Geometric Constraints:** The point x creates two segments: x and $L - x$. The ratio R is $\frac{\min(x, L-x)}{\max(x, L-x)}$.
- **Case Analysis:** We must split the analysis at the midpoint $L/2$ to determine which segment is shorter.

Step-by-Step Solution

- **Case 1 ($x \leq L/2$):** Shorter segment is x; longer is $L - x$.

$$\frac{x}{L-x} < \frac{1}{4} \Rightarrow 4x < L - x \Rightarrow 5x < L \Rightarrow x < \frac{L}{5}$$

- **Case 2 ($x > L/2$):** Shorter segment is $L - x$; longer is x.

$$\frac{L-x}{x} < \frac{1}{4} \Rightarrow 4L - 4x < x \Rightarrow 5x > 4L \Rightarrow x > \frac{4L}{5}$$

- **Probability Calculation:** Valid regions are $(0, L/5)$ and $(4L/5, L)$. Total valid length $= (L/5 - 0) + (L - 4L/5) = 2L/5$.

$$P = \frac{\text{valid region}}{\text{total region}} = \frac{2L/5}{L} = \frac{2}{5}$$

Q2. Bayesian Inference: Image Classification

Problem: A point with reading $X = 5$ is chosen from an image with black (fraction α) and white (fraction $1-\alpha$) regions. Find α such that the probability of error is the same for either classification.

Reasoning & Logic

- **Equal Error Condition:** For the error probability to be identical regardless of the conclusion, the posterior probabilities must be equal: $P(B|X = 5) = P(W|X = 5) = 1/2$.
- **Bayes' Theorem:** We use the Likelihoods $f(x|B)$ and $f(x|W)$ along with Priors $P(B)$ and $P(W)$ to find the posterior.

Step-by-Step Solution

- **Distributions:** $X|W \sim N(4, 4) \Rightarrow \sigma_1 = 2$; $X|B \sim N(6, 9) \Rightarrow \sigma_2 = 3$.
- **Normal Density Formula:** $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$.
- **Likelihoods at $X = 5$:**

$$f(5|B) = \frac{1}{3\sqrt{2\pi}} e^{-1/18}; \quad f(5|W) = \frac{1}{2\sqrt{2\pi}} e^{-1/8}$$

- **Equating Posteriors:**

$$\frac{\alpha f(5|B)}{\alpha f(5|B) + (1-\alpha) f(5|W)} = \frac{1}{2} \Rightarrow \frac{3\alpha}{3\alpha + 2(1-\alpha)e^{-5/72}} = \frac{1}{2}$$

- **Final Result:** $3\alpha = 2(1-\alpha)(0.93) \Rightarrow 4.86\alpha = 1.86 \Rightarrow \alpha \approx 0.38$.

Q3. Optimization: Fire Station Location

Problem: Minimize $\mathbb{E}[|X - a|]$ for (a) $X \sim \text{Uniform}(0, A)$ and (b) $X \sim \text{Exponential}(\lambda)$.

Reasoning & Logic

To minimize the expected absolute distance, we define the expectation as an integral, split at the point of the absolute value change (a), and differentiate with respect to a.

Step-by-Step Solution

- **(a) Uniform Case:** $\mathbb{E}[|X - a|] = \int_0^a (a - x)\frac{1}{A}dx + \int_a^A (x - a)\frac{1}{A}dx$. Differentiating and setting to zero:

$$\frac{d}{da}\left(\frac{2a^2 - 2aA + A^2}{2A}\right) = \frac{1}{2A}(4a - 2A) = 0 \Rightarrow \textbf{Result: } a = A/2 \text{ (midpoint)}$$

- **(b) Exponential Case:** $\mathbb{E}[|X-a|] = \int_0^a (a-x)\lambda e^{-\lambda x}dx + \int_a^\infty (x-a)\lambda e^{-\lambda x}dx$. Differentiating w.r.t a:

$$1 - 2e^{-\lambda a} = 0 \Rightarrow e^{-\lambda a} = 1/2 \Rightarrow \textbf{Result: } a = \frac{\ln 2}{\lambda}$$

Q4. Conditional Survival: Mixed Batteries

Problem: Find $P(X > t + s | X > t)$ for a mixture of two exponential battery types.

Reasoning & Logic

- **Law of Total Probability:** Used to find $P(X > t)$ by summing the survival probabilities of each type weighted by their selection probability.
- **Conditional Probability:** $P(A|B) = P(A \cap B)/P(B)$. For survival, $P(X > t + s \cap X > t) = P(X > t + s)$.

Step-by-Step Solution

- $P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$.
- $P(X > t + s) = p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}$.
- **Result:** $P(X > t + s | X > t) = \frac{p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}$.

Q5. Conditional Poisson Requests

Problem: Given $X \sim \text{Poi}(\lambda_1)$ and $Y \sim \text{Poi}(\lambda_2)$ are independent, find $P(X = k | X + Y = n)$.

Reasoning & Logic

The sum of independent Poisson variables is also Poisson with rate $\lambda_1 + \lambda_2$. We seek the ratio of the joint probability $P(X = k, Y = n - k)$ to the total probability $P(X + Y = n)$.

Step-by-Step Solution

- $P(X = k | X + Y = n) = \frac{P(X=k)P(Y=n-k)}{P(X+Y=n)}$.
- Substitute PMFs: $\frac{\left(\frac{e^{-\lambda_1}\lambda_1^k}{k!}\right)\left(\frac{e^{-\lambda_2}\lambda_2^{n-k}}{(n-k)!}\right)}{\frac{e^{-(\lambda_1+\lambda_2)}(\lambda_1+\lambda_2)^n}{n!}}$.
- **Result:** $\binom{n}{k}\left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1+\lambda_2}\right)^{n-k}$. (This is the Binomial distribution).

Q6. Transformation: Mixed Distribution

Problem: $X \sim \text{Uniform}[-2, 2]$. $Y = g(X)$ where $g(x) = x$ if $|x| \geq 1$, and $g(x) = 0$ if $|x| < 1$.

Reasoning & Logic

This transformation is a "crushing" function where a range of X values map to a single value $Y = 0$. This creates a point mass (discrete component) in the CDF.

Step-by-Step Solution

CDF $F_Y(y)$:

- $y \in [-2, -1) : F_Y(y) = \frac{y+2}{4}$.
- At $y = 0 : P(Y = 0) = P(-1 < X < 1) = 2/4 = 1/2$.
- $F_Y(0) = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$.
- $y \in [1, 2] : F_Y(y) = \frac{y+2}{4}$.

PDF $f_Y(y)$: $1/4$ for $y \in (-2, -1) \cup (1, 2)$, with point mass $P(Y = 0) = 1/2$.

Q7. Stock Movement Approximation

Problem: After 1000 periods, find the probability stock price is up $\geq 30\%$ given $u = 1.012, d = 0.990, p = 0.52$.

Reasoning & Logic

The price $S_{1000} = S_0 u^k d^{1000-k}$, where k is the number of up-moves. We take natural logs to solve for the required k.

Step-by-Step Solution

- $u^k d^{1000-k} \geq 1.30 \Rightarrow k(\ln u - \ln d) + 1000 \ln d \geq \ln(1.30)$.
- $k \geq \frac{\ln(1.30) - 1000 \ln d}{\ln u - \ln d} \approx 469.4 \Rightarrow k \geq 470$.
- **Result:** $k \sim \text{Binomial}(1000, 0.52)$. $P(k \geq 470) = \sum_{k=470}^{1000} \binom{1000}{k} (0.52)^k (0.48)^{1000-k}$.

Q8. Exponential Repair Times

Problem: $X \sim \text{Exp}(1/2)$. Find (a) $P(X > 2)$ and (b) $P(X \geq 10 | X > 9)$.

Step-by-Step Solution

- **(a):** $P(X > 2) = \int_2^\infty \frac{1}{2} e^{-x/2} dx = e^{-1} \approx 0.3679$.
- **(b):** $P(X \geq 10 | X > 9) = \frac{P(X \geq 10)}{P(X > 9)} = \frac{e^{-5}}{e^{-9/2}} = e^{-1/2}$.

Q9. PDF of $Z = \sqrt{X^2 + Y^2}$

Problem: Find the PDF of Z for independent X, Y.

Reasoning & Logic

The CDF $P(Z \leq z)$ represents the joint density integrated over a circular region $x^2 + y^2 \leq z^2$. We use **Leibniz's Rule** to differentiate the integral with respect to its limits.

Step-by-Step Solution

- $F_Z(z) = \int_{-z}^{z} \int_{-\sqrt{z^2-x^2}}^{\sqrt{z^2-x^2}} f_X(x) f_Y(y) dy dx$.
- Differentiating w.r.t. z using Leibniz's rule: $\frac{d}{dz}\sqrt{z^2 - x^2} = \frac{z}{\sqrt{z^2-x^2}}$.
- **Result:** $f_Z(z) = z \int_{-z}^{z} \frac{f_X(x)[f_Y(\sqrt{z^2-x^2}) + f_Y(-\sqrt{z^2-x^2})]}{\sqrt{z^2-x^2}} dx, z > 0$.

Q10. Joint Distribution of Maxima

Problem: Find the joint CDF of $M_n = \max(X_1, \ldots, X_n)$ and M_{n+1}.

Reasoning & Logic

$M_{n+1} = \max(M_n, X_{n+1})$. We analyze the joint event $P(M_n \leq x, M_{n+1} \leq y)$ based on the relative values of x and y.

Step-by-Step Solution

- **Case 1 ($x \leq y$):** Requires $X_1, \ldots, X_n \leq x$ AND $X_{n+1} \leq y$.
$$P = P(M_n \leq x) P(X_{n+1} \leq y) = F(x)^n F(y)$$
- **Case 2 ($x > y$):** If $M_{n+1} \leq y$, then M_n is automatically $\leq y$. Since $y < x$, the condition $M_n \leq x$ is redundant.
$$P = P(M_{n+1} \leq y) = F(y)^{n+1}$$
- **Final Joint CDF:**
$$\begin{cases} F(x)^n F(y) & x \leq y \\ F(y)^{n+1} & x > y \end{cases}$$